

Distributed Computing and its Links to Quantum Information Theory and the NPA Hierarchy

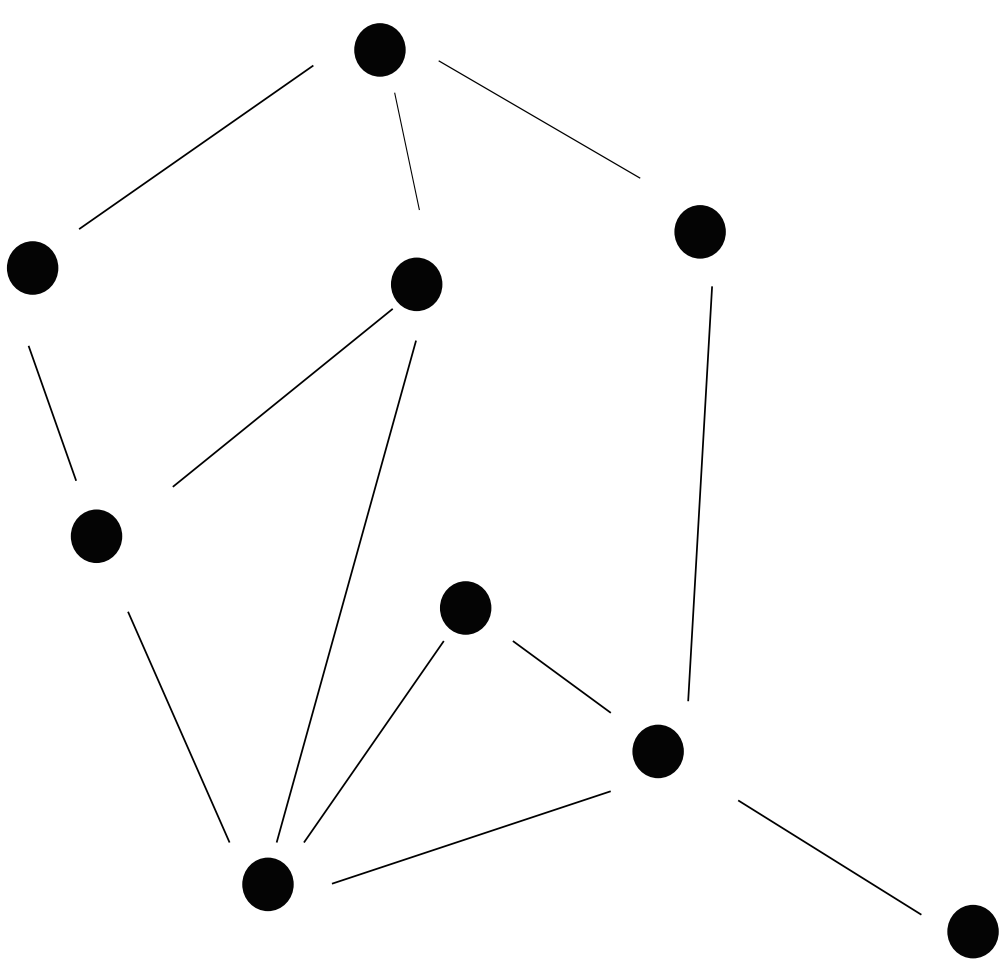
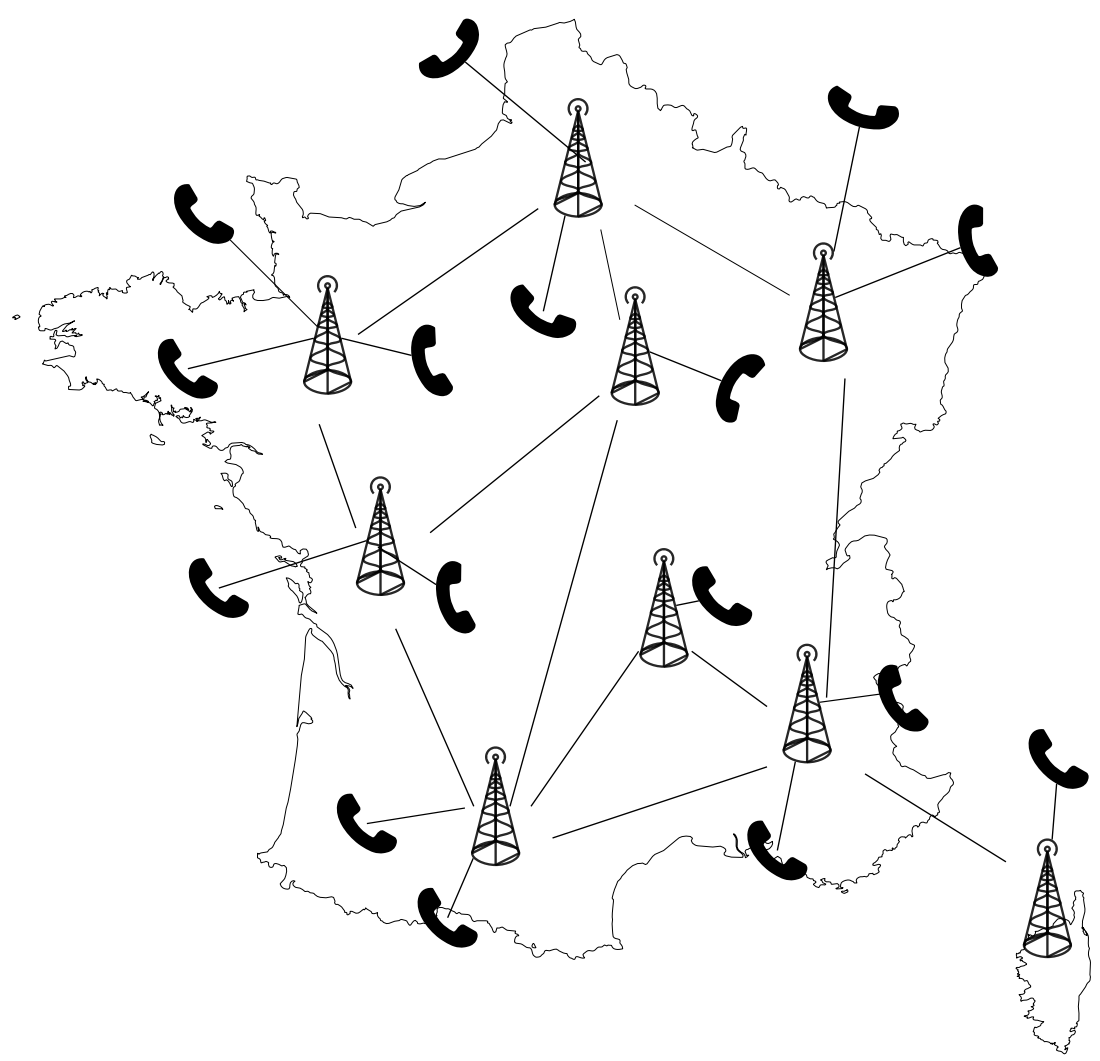


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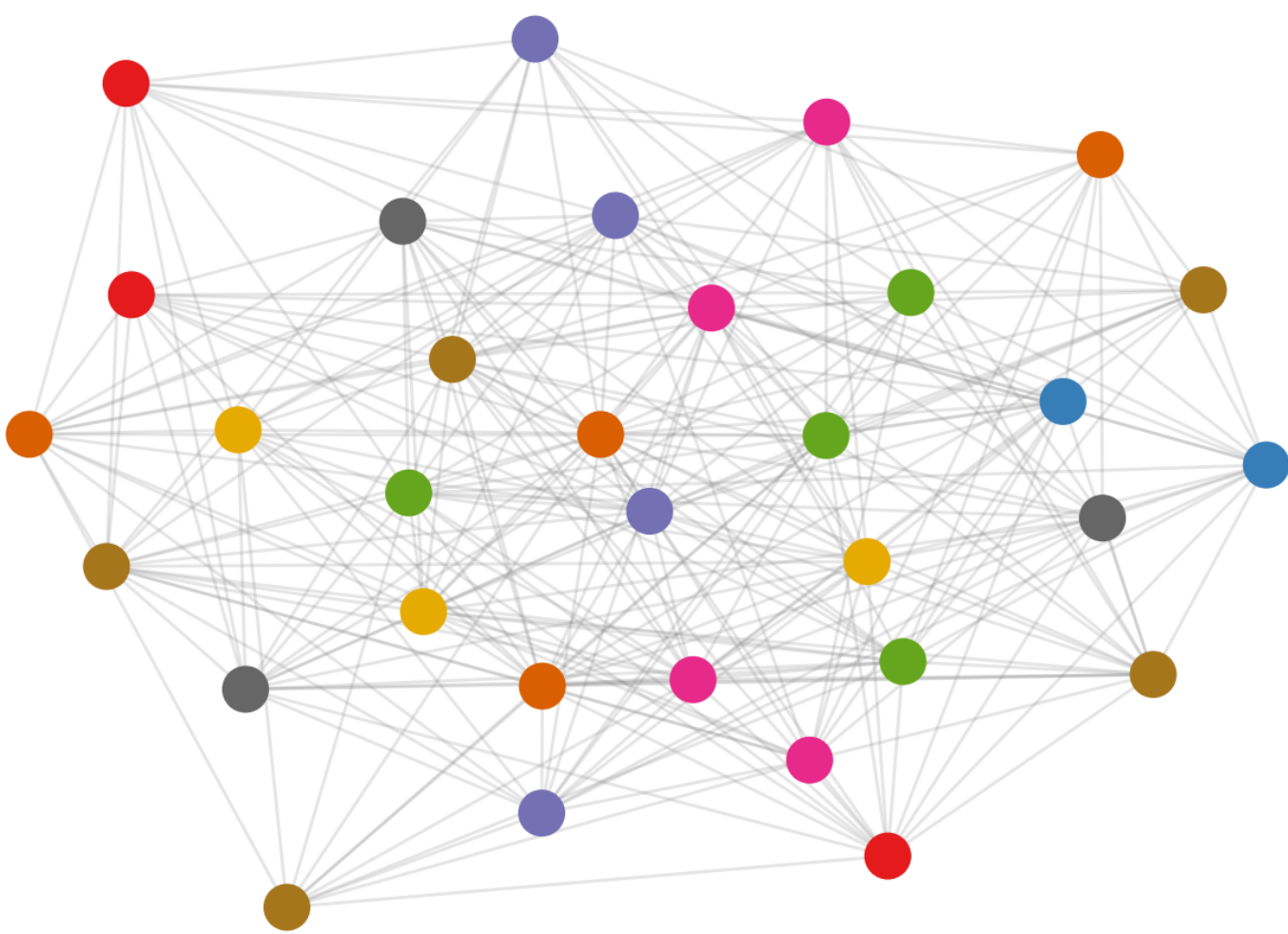
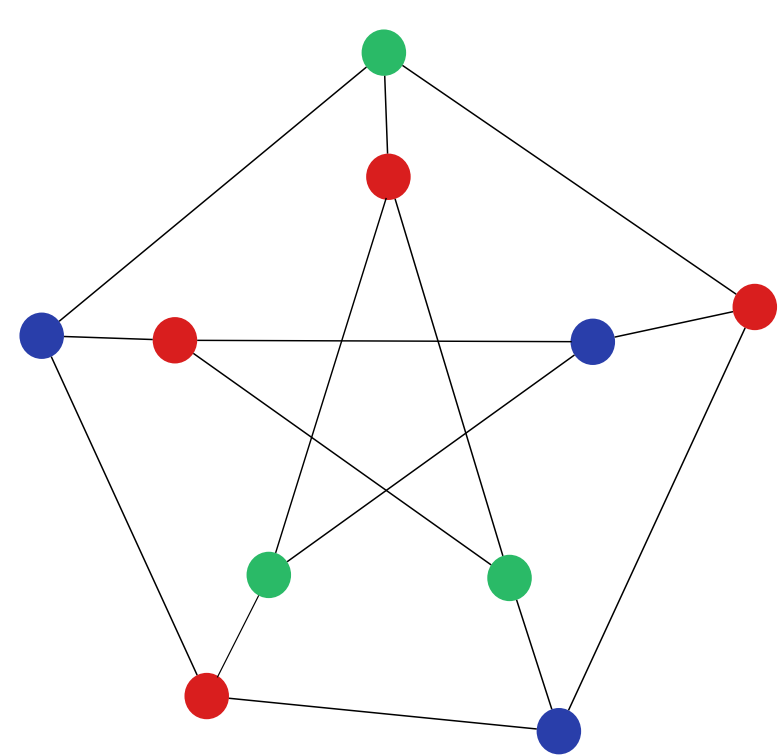
What is Distributed Computing?



Goal: Collaboratively solve a distributed task in a network that involves multiple identically operating computers that communicate with each other.

- Computers and communication links are organized according to graph structure.
- Exchange of information according to synchronized clock.
- No centralized administrator $\Rightarrow$ **Only local information available.**
- Channels have unlimited capacity.
- Unlimited local computational power $\Rightarrow$ **Complexity due to geometry of the problem.**

The Canonical Graph Problem: Coloring



- No neighboring nodes can output the same color $\Rightarrow$ **Locally Checkable Labeling (LCL) Problem.**
- Easy to solve for a centralized administrator.
- At the basis of various other distributed tasks.

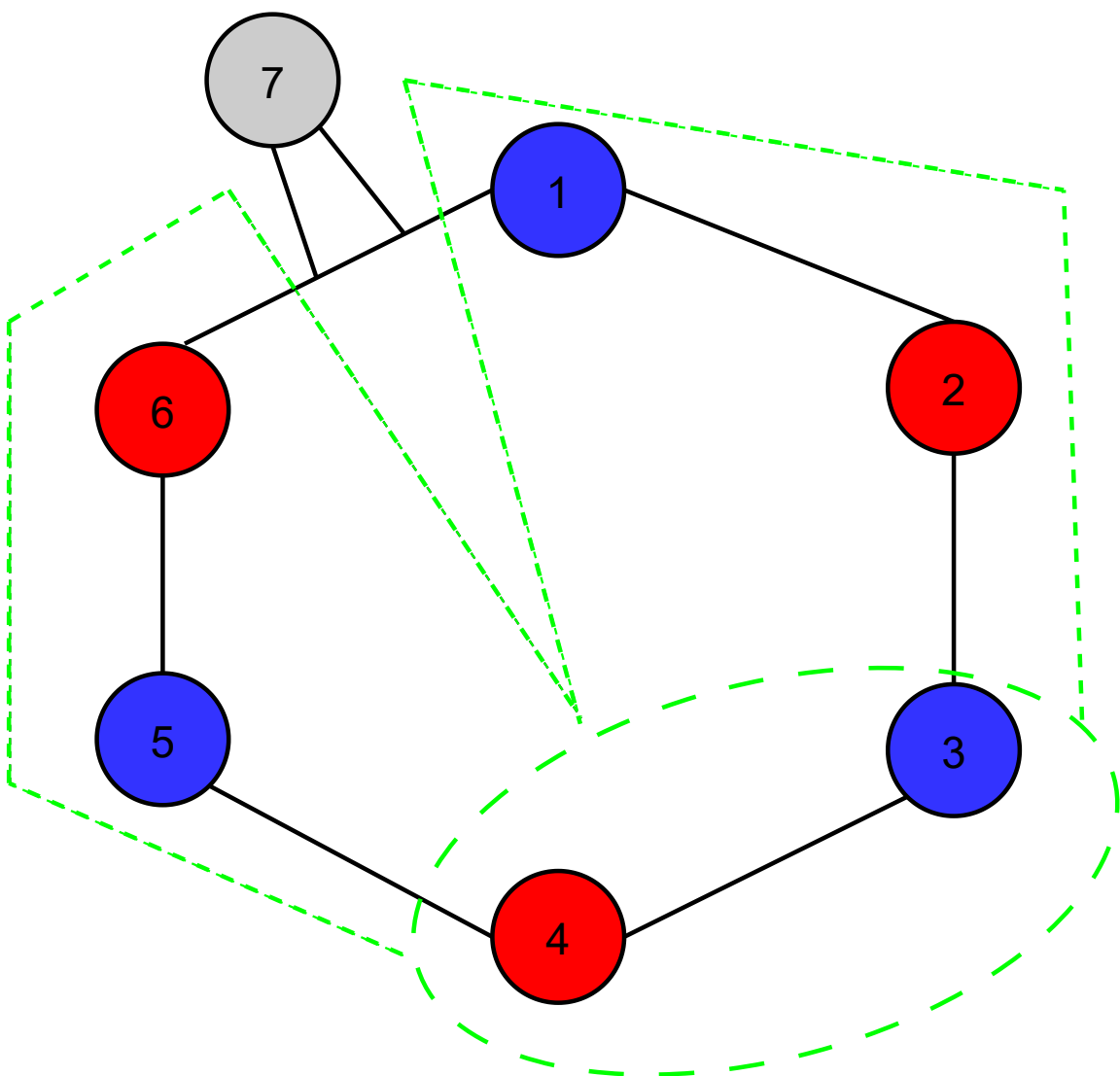
Figure of merit:

How many rounds T of synchronized communication are necessary to solve the problem?

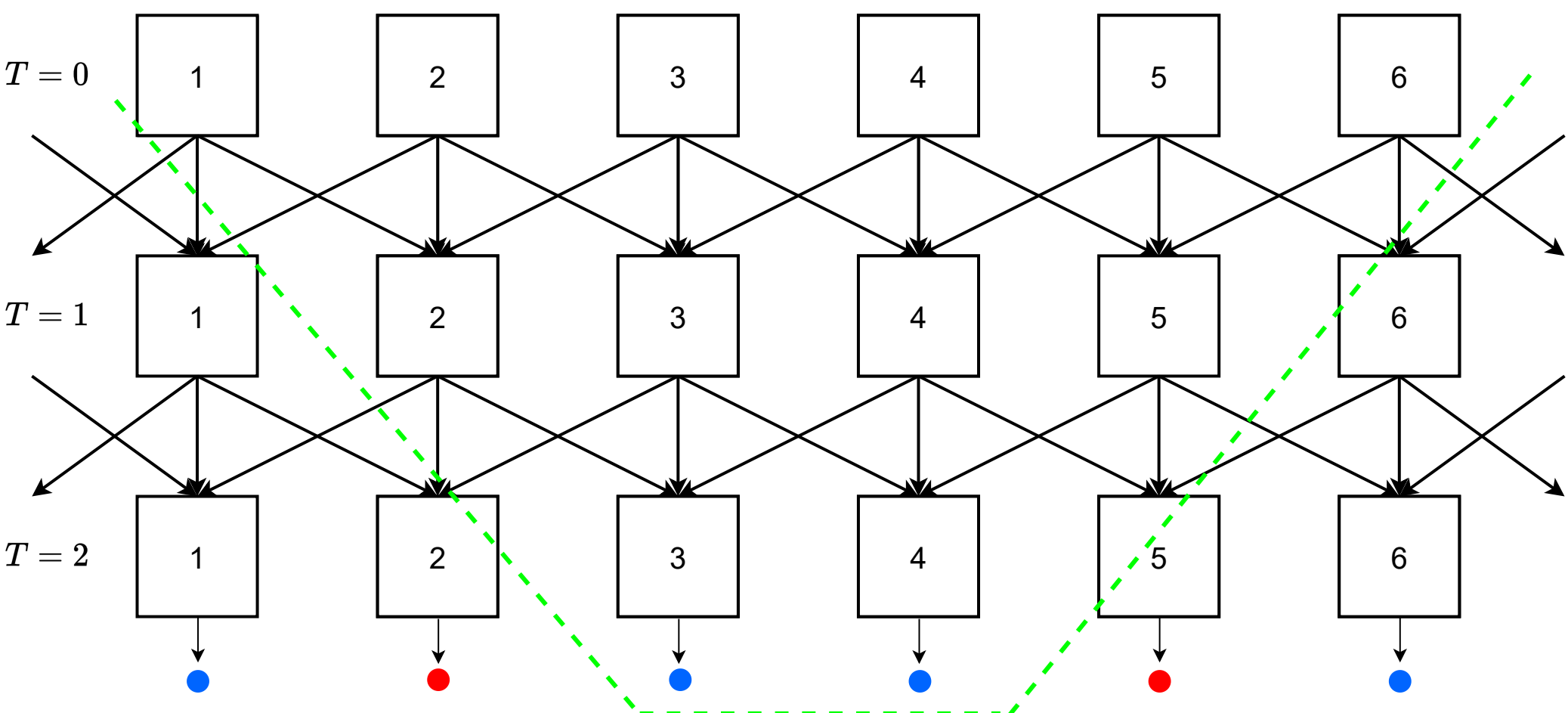
- Interest in solution for family of graphs with scaling parameter n .
- Standard problem in classical distributed computing.
- Different models of computation $\Rightarrow$ **Focus here on the LOCAL model.**
- Different underlying physical theories: Classical, Quantum, No-Signaling.

Distributed Quantum Computing and a First Example

Task: Color a 6-Loop using two colors with as few communication rounds T as possible.



- In $T = 3$ steps all information from neighbors can be collected.
- A leader node can decide the output of each node.
- Leader node can be decided retrospectively by drawing random (real) numbers.
- $T \leq 2$ steps can be ruled out through no-signaling and device replication.
- The behavior of the sub-graph $(3, 4)$ after $T = 2$ steps is independent of the existence of node 7.



- Nodes can locally produce any entangled state and perform any quantum measurement.
- Entangled states can be distributed via quantum channels to neighboring nodes.
- Any quantum channel can be applied locally.

The State of the Art

A no-go result^[1]:

- Consider a graph G of n nodes.
- Let χ be the chromatic number of G .
- Try to color G with $c > \chi$ colors.

$\Rightarrow$ **No quantum advantage!**

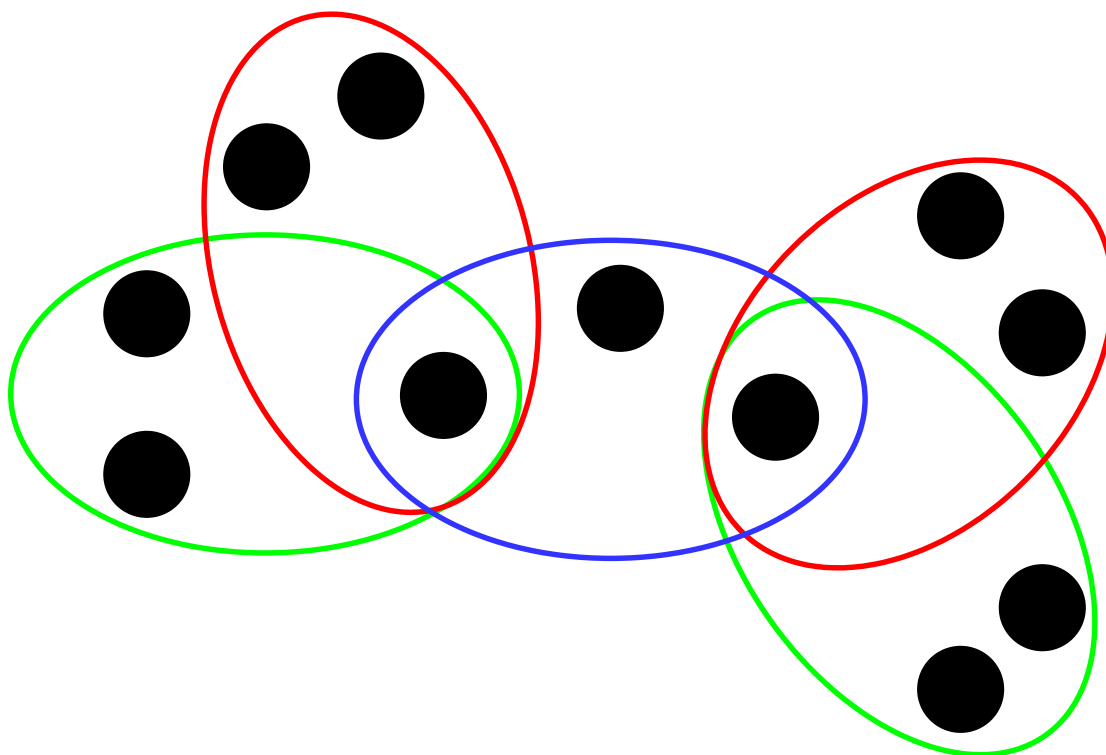
Idea: Sandwich quantum run time T_Q between best known classical time T_C and no-signaling lower bound T_{NS} . Find $T_C \approx T_{NS}$.

A first quantum advantage^[2]:

- Consider a regular hyper-graph tree of degree Δ .
- Play the GHZ game iteratively on hyper-edges.
- Outputs of round $(n - 1)$ become inputs of round n .

$\Rightarrow$ **The game can be won deterministically by distributing GHZ states in 1 round.**

Classically: $\Omega(\Delta)$ rounds of communication are necessary.



Overview of the loop coloring complexity

Theory \ Colors c	2	3	$c \geq 4$
Classical	$T \sim n/2$	$T \sim \log^*(n)$	$T \sim \log^*(n)$
Quantum	$T \sim n/2$	???	???
No-signaling	$T \sim n/2$	$T = 2$	$T = 1$

Table 1: Number of communication steps T needed with c colors in a given theory^[3,4].

How to Bound the Capabilities of Quantum Theory?

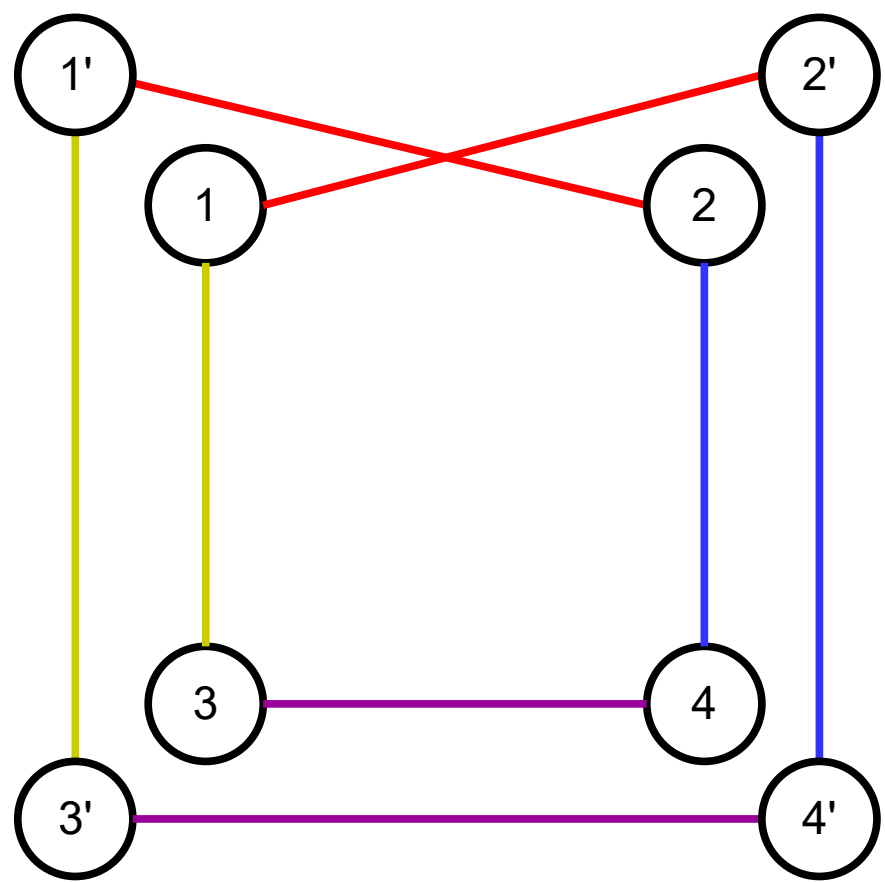
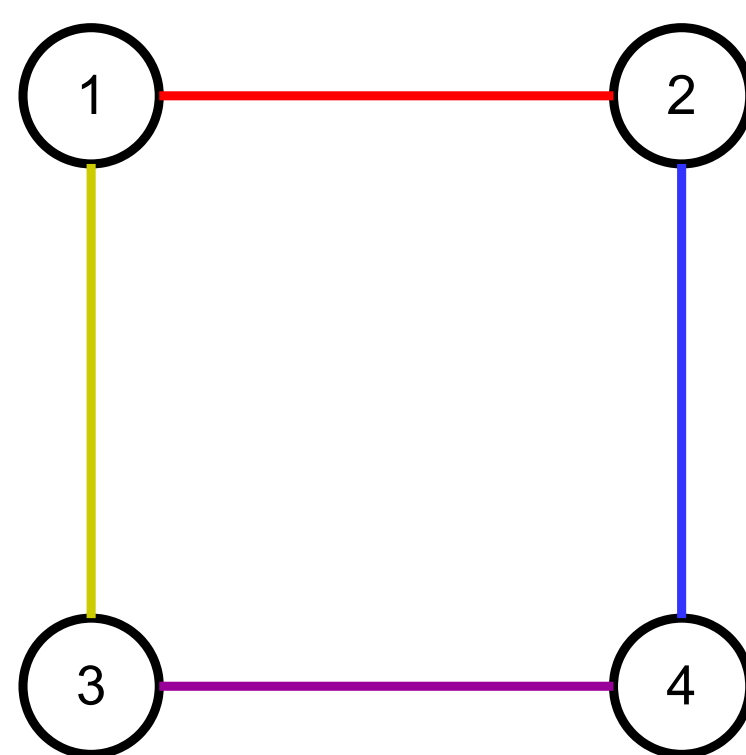
Main Question:

How to lower bound the needed communication steps T_Q without relying on no-signaling arguments?

$\Rightarrow$ **Which correlations $p(a_1, a_2, \dots, a_{n-1}, a_n)$ can be achieved by n quantum nodes in $T \ll n$ steps?**

Bounding classical strategies: Method of round elimination^[5] $\Rightarrow$ Relation to fan-out inflations.

Inflation



$\Rightarrow$ **Non-trivial constraints arise due to symmetries.**

NPA Hierarchy + LCL problems?

- The NPA Hierarchy can bound quantum correlations in Bell scenarios.
- Combining NPA+Inflation allows us to study quantum network correlations.
- Asymptotic statements in n are necessary.
- Our problems should be **LCL**, how to work with a fixed locality radius?

Including Communication

- How to capture and mathematically describe the communications steps?
- Adapt graph structure for every step of communication?
- Can we have an "NPA hierarchy for distributed computing" including communication?

Conclusions

- First demonstration of a quantum advantage in distributed computing for **LCL** problems.
- Methods to bound classical strategies exists, close to fan-out inflation techniques.
- So far: No-signaling arguments are used to bound quantum strategies.
- Coloring of a loop demonstrates the need for more elaborate techniques tailored to the quantum formalism.

References

- [1] X. Coiteux-Roy *et al.*, Proceedings of STOC '24, **1901-1910**.
- [2] A. Balliu *et al.*, arXiv:2411.03240 (accepted for Proceedings of STOC '25).
- [3] A. Holroyd, T.M. Liggett, arXiv:1403.2448.
- [4] J. Rybicki, arXiv:0901.4709 (2009).
- [5] S. Brandt, Proceedings of PODC '19, **379 - 388**.

Credits to Isadora Veeren for creating multiple figures.